

MGTA 454 Business Analytics Capstone

Spring 2026

OFFICE HOURS: By appointment; please reach out to faculty advisor or teaching assistant on email
Team meetings will be scheduled individually.

DESCRIPTION

In today's digital age, leveraging data-driven insights is essential for achieving a competitive edge. Companies that harness the power of data continuously develop new methods, algorithms, and tools to address complex business challenges.

This capstone course offers you the opportunity to apply the strategies, techniques, and tools learned throughout the MSBA program to solve real-world business and analytics problems. Working in teams on a project sponsored by an external organization, you will engage in every step of the data analytics process—from defining the problem and preparing data to analysis and final presentation. This course is designed to prepare you for a successful career in analytics by providing practical, hands-on experience in teamwork, project management, and professional communication.

It is essential that every team member contributes their data analysis skills and business acumen to achieve the project objectives.

OBJECTIVES

By the end of this course, you will be able to:

- Experience the full cycle of designing and implementing analytical solutions for practical business problems.
 - Negotiating the scope of the project
 - Prioritizing certain parts of the project in case the scope of work is greater than the time/resources available
 - Working with peers from different cultures and different skill-sets
 - Replicating a real-life situation where skills needed for a project often change very often and at short notice - learning on the job, where the technical skill-set needed for the project is not something you have acquired so far, but you acquire those skills while executing the project
- Engage with real clients to address authentic business challenges.
 - Asking for help and support from relevant person if you are running behind schedule
 - Communicating the progress (or lack of it) clearly to all stakeholders
 - Managing stakeholder expectations
 - Communicating your intermediate and final output in a manner which is understandable to all stakeholders and persuades them to agree to your recommendations

PROJECTS

Your capstone project, sponsored by an external company or organization, will include the following activities:

1. Create a Project Charter and get it approved by the client
2. Collaborate with clients to clearly define a business problem and design an analytics solution.
3. Review relevant technical literature and consult with subject matter experts.
4. Organize, cleanse, and preprocess the available data.
5. Perform exploratory data analysis to uncover insights.
6. Update the Project Health Tracker with the latest status every week
7. Conduct predictive data analysis to model future trends.
8. Prepare a comprehensive data product, which may include visualizations, code/scripting, and

even app development in certain cases.

9. Communicate your key findings and insights through presentations.

REPOSITORIES AND COURSE MANAGEMENT TOOLS

Each project team will have a private repository available on the MSBA server, through the UCSD supercomputing center, or computing resources provided by their sponsor organization. Access instructions will be provided as needed. Communication with the class and within teams will primarily be conducted through Canvas, Zoom, Google Docs and email. Here are the tools you will use throughout the course:

- EduSourced: For bidding on project proposals
- Canvas: For assignment submissions, announcements, scheduling, and recording presentations.
- Google Sheets: Project Health Tracker (individual tracker for each team)
- Executive summary slide ([link to template – use any one of the eight slides](#))
- Google docs: Project charter template ([link to template](#))
- Peer Feedback Survey tool: To be used every 2 weeks to share peer feedback; you should expect a link to the survey in your mailbox every 2 weeks.

Important note: No data to be stored on your personal laptops or personal storage devices.

STUDENT WORKLOAD

MGTA 454 is a 4-credit course, which typically requires an average commitment of 12 hours per week.

GRADING ([Link to detailed rubric](#))

1. Overall Grading for the course: Presentations ONLY form one of the areas on which grades will be based. Here is the link to the overall grading rubric:

<https://docs.google.com/document/d/1vzfJX693VP8at5joEEruVAF2HLXqa0I/edit?usp=sharing&oid=101911717995615379677&rtpof=true&sd=true>

2. Detailed grading rubric for mid-term presentation:

<https://docs.google.com/spreadsheets/d/1Lr-A5q1KkZ15uhwCEIFQQvQARXeAYFL7UE--ubkpHJg/edit?gid=1336093996#gid=1336093996>

3. Detailed grading rubric for final presentation:

https://docs.google.com/spreadsheets/d/1wXl6WaQpu80hqy_jhkJJ0l6WK99ahIJ7R1YY2s3N-Ws/edit?gid=1336093996#gid=1336093996

You will be evaluated based on your individual performance as well as your team's work products, based on the ability to demonstrate the following:

1. **Professionalism (25%) – Individual rating**
 - Preparedness in meeting with faculty advisor: Are you on time and are you ready to present what you had committed
 - Preparedness in meeting with client: Are you on time and are you ready to present what you have committed
 - Diligence: Ability to deliver every week on your committed deliverables
2. **Team-work (25%) – Individual rating**
 - A peer feedback survey conducted every 2 weeks where you will be asked to evaluate the performance and professionalism of your team-mates
 - Filling the peer feedback survey is mandatory to get the grades for the peer feedback
3. **Quality of output (15%) – Group rating**
 - Ability to define and operationalize key questions and data analytic deliverables.
 - Design a solution path that connects business challenges with data analytics strategies.

- Responsiveness: Does the team respond to emails from clients and faculty advisors within 24 hours
- Develop and execute a comprehensive data analysis project.
- Clearly articulate the analytical methods used to tackle the business problem.
- 4. **Communication skills (15%) – Individual rating**
 - Communicating effectively with business, technical, and academic audiences
 - Keeping faculty advisor, client and your team members aware of any potential delays or issues
- 5. **Understanding of final output (20%) – Individual rating**
 - Even though you may not have worked on the entire project, you should be able to explain the inputs and outputs clearly with very little assistance from your team members

PEER GRADING (Team-work)

- Peer evaluation forms will need to be filled by you every 2 weeks for a total of 5 times
- These evaluations provide feedback on each team member's contributions and will affect your overall participation grade.
- Evaluations are confidential

SCHEDULE

Notes:

- Please invite Anshul Sachdev (a1sachdev@ucsd.edu) for all meetings (virtual or in-person) with your faculty advisor and company representatives.
- When setting up a virtual meeting, please use a Zoom calendar invite. Do NOT send the zoom link in an email. Sending Zoom links in email does not appear in the calendar of the person you want to meet, and it does not remind the person of the meeting. Calendar invites improve attendance, reduce scheduling conflicts, and provide automated reminders.
- Actual meeting dates between students and company representative(s) would depend on mutual agreement between them. Frequency of meetings between company representative(s) and student team will be at least once per week.
- Actual meeting dates between students and faculty advisors would depend on mutual agreement between them. Frequency of meetings between faculty advisor and student team will be at least once per week. Faculty Advisor may or may not join client meetings.

Week	By this date	Discussion and Tasks
0	3/30	Project kickoff meeting with company representative (exact date to be determined based on client availability but before March 30) – project scoping, success criteria and setting up a recurring weekly meeting schedule. Set up a recurring meeting schedule with your faculty advisor too.
1	4/3	Internal meeting - Group progress report and discussion with faculty advisor. Set recurring Share project charter with roles/responsibilities, project plans and milestones with faculty advisor and client. Get a sign-off on the project charter from the client.
2	4/10	Meet company representatives to provide updates, ask for business context, discuss any challenges and follow-up on any pending action (data sharing etc.) Internal meeting - Group progress report and discussion with faculty advisor. Submit Executive Summary slide and Project Charter on Canvas. Team lead to update

		Project Health Tracker for the team. Team lead to update Project Health Tracker for the team and team members to update Project Health Tracker for their respective tasks and completion status. Each team member fills the peer review survey. It is mandatory to fill this survey and not optional.
3	4/17	Meet company representatives to provide updates, ask for business context, discuss any challenges and follow-up on any pending action (data sharing etc.) Group progress report and discussion with faculty advisor; Submit updated Executive Summary slide. Team lead to update Project Health Tracker for the team and team members to update Project Health Tracker for their respective tasks and completion status.
4	4/24	Start putting the finishing touches on the mid-term presentation. Meet company representatives to provide updates, ask for business context, discuss any challenges and follow-up on any pending action (data sharing etc.) Group progress report and discussion with faculty advisor; share updated Executive Summary slide and draft mid-term presentation with faculty advisor and get their feedback. Team lead to update Project Health Tracker for the team and team members to update Project Health Tracker for their respective tasks and completion status. Each team member fills the peer review survey. It is mandatory to fill this survey and not optional. Set up time with your communications coach to meet them twice before the mid-term presentation.
5	5/1	Present your PowerPoint slide deck to your communication coach for an initial review. Incorporate their inputs in the presentation. Meet company representatives to provide updates, ask for business context, discuss any challenges and follow-up on any pending action (data sharing etc.) Group progress report and discussion with faculty advisor; Team lead to update Project Health Tracker for the team and team members to update Project Health Tracker for their respective tasks and completion status.
6	5/5	Meet your communication coach to show them the presentation after incorporating their inputs. Mid-term Presentation & Exam; group presentation and Q&A followed by Individual Q&A. Team lead to update Project Health Tracker for the team and team members to update Project Health Tracker for their respective tasks and completion status. Each team member fills the peer review survey. It is mandatory to fill this survey and not optional. Submit updated Executive Summary slide and mid-term presentation (both in PowerPoint format; NOT pdf) and Project Charter (word document format). Meet company representatives to provide updates, ask for business context, discuss any challenges and follow-up on any pending action (data sharing etc.)
7	5/15	Meet company representatives to provide updates, ask for business context, discuss any challenges and follow-up on any pending action (data sharing etc.) Group progress report and discussion with faculty advisor. Team lead to update Project Health Tracker for the team and team members to update Project Health Tracker for their respective tasks and completion status.
8	5/22	Email the latest version of your slide deck to Anshul Sachdev (a1sachdev@ucsd.edu) to be considered for The Rady Analytics Capstone

		Excellence (TRACE) Awards. The subject line of your email should say "TRACE" Set up time with your communications coach to meet them twice before the mid-term presentation. Complete Final presentation slides and meet Faculty advisor for feedback. Group progress report and discussion with faculty advisor. Team lead to update Project Health Tracker for the team and team members to update Project Health Tracker for their respective tasks and completion status. Each team member fills the peer review survey. It is mandatory to fill this survey and not optional.
9	5/29	Present your PowerPoint slide deck to your communication coach for an initial review. Incorporate their inputs in the presentation. Final Presentation & Exam; group presentation and Q&A followed by Individual Q&A; Team lead to update Project Health Tracker for the team and team members to update Project Health Tracker for their respective tasks and completion status.
10	6/5	Meet your communication coach to show them the presentation after incorporating their inputs. Presentations to client leadership based on client availability; in-person/virtual depending on client availability. Submit updated Executive Summary slide and Final presentation (both in PowerPoint format; NOT pdf) and Project Charter (word document format). Team lead to update Project Health Tracker for the team and team members to update Project Health Tracker for their respective tasks and completion status. Each team member fills the peer review survey. It is mandatory to fill this survey and not optional.

GROUP PRESENTATIONS

- **Mid-Term Presentation:**
 - A concise 15-minute presentation with a maximum of 12 slides in the main deck, outlining key aspects of your capstone project; you can have additional slides in the appendix to answer any questions from the audience
 - Suggested content includes:
 - A description of the business problem and any constraints.
 - The business goal and available data.
 - Expected deliverables and metrics for success.
 - Your proposed solution path and an overview of data preparation, exploratory analysis, and predictive or explanatory analysis.
 - Initial findings, recommendations, and the potential commercial value of your results.
 - 30 minutes - Feedback & faculty coaching to address any gaps or blind spots
- **Final Presentation:**
 - A 15-minute presentation with a maximum of 12 slides in the main deck, showcasing your final results and recommendations; you can have additional slides in the appendix to answer any questions from the audience
 - Your presentation should include:
 - Clear explanation of your predictive model, error metrics, and evaluation methods.
 - Detailed discussion of your prescriptive recommendations; how they create business value.
 - An interactive Q&A session where you refine your project based on audience feedback.
 - Q&A for 30 minutes (includes team & Individual)

Each presentation should be clear, concise, and designed to engage a nontechnical audience.

Mandatory presentation style

You can add client logos to the presentation but you have to use the slide style, format, color scheme and templates below:

1. Flow of presentation:

<https://docs.google.com/document/d/1kDbAWfTuaT2GBcinYXIDfDraAMMjkEJb/edit?usp=sharing&ouid=101911717995615379677&rtpof=true&sd=true>

2. Style and color scheme of presentation: Make a copy of this sample presentation to kick-start your presentation -

https://docs.google.com/presentation/d/14GcW2rLLNhZf5D_aYUDfF8X9gkZ0gaIF/edit?usp=sharing&ouid=101911717995615379677&rtpof=true&sd=true

3. Repository of sample visuals:

https://docs.google.com/presentation/d/1V7sHXgd69EFb9jRebGv2Y8gBm8f0DKE0/edit?usp=drive_link&ouid=101911717995615379677&rtpof=true&sd=true

Note: Presentations ONLY form one of the areas on which grades will be based. Please see section on Grading to understand the various components of your grade.

WEEKLY REVIEW MEETINGS

- Teams are required to hold weekly meetings with their faculty supervisor and with the company representatives, and every team member must attend.
- If you cannot attend a meeting for a valid reason, notify your team and the supervisor in writing at least one day in advance.
- Meetings should begin with a review of the previous week's progress followed by planning for the upcoming week.
- Regular face-to-face meetings with full team attendance are strongly encouraged for success.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at: <http://www-senate.ucsd.edu/manual/appendices/app2.htm#AP14>

USE OF AI TOOLS

Use of AI tools such as ChatGPT is permitted in this class for the purposes of enhancing learning, however, it is the responsibility of each student to ensure that their use of AI tools does not constitute a, academic misconduct such as plagiarism or cheating, or non-compliance against company sponsored projects that contain non-disclosure/confidentiality agreements. This includes the proper handling of data, code, and company information that may be considered protected, regulated, or contain trade secrets. Use caution and discretion when using external tools like ChatGPT and ensure you are informed and up to date with university and sponsor company policies before using these tools to avoid any breach of security or compliance with any confidentiality agreements.

HOW THE HONOR CODE APPLIES TO THIS CLASS

This course will prepare you for professional positions in the analytics field. There you will be expected to work effectively in teams, where complex projects are divided into tasks to be completed by individual members. In this environment every person's contribution is expected and explicitly known. In your team project it is expected that each member contributes substantively to the final product, as is expected in an actual work situation.

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. No accommodations can be implemented retroactively.

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.